Colon & Colorectal Cancer — Biomarker Testing Intelligence

Alert Window: Last 30 Days · Generated: 2026-10-11 07:30:01

Situation Summary

Stage IV: 5 of 36
Biomarker Count: 21 entries
Top Targets: ctDNA (6) · NGS (5) · MSI/MMR (5)
Breakthrough: 10 entries scored ≥ 4

Research & Guidelines Timeline

Biomarker testing entries ranked by significance score. Higher scores indicate greater clinical relevance. Always discuss with your oncologist.

HIGH Stage IV ctDNA VERIFIED LIQUID BIOPSY PUBLISHED SEP 25, 2026 11

Circulating tumor DNA tumor fraction as a biomarker in refractory metastatic colorectal cancer: findings from the CAVE-2 GOIM trial

In the CAVE-2 GOIM trial, circulating tumor DNA (ctDNA) was assessed using the FoundationOne Liquid assay to evaluate its role in guiding anti-EGFR rechallenge therapy with cetuximab ± avelumab in refractory metastatic colorectal cancer patients. The study highlights the need for biomarkers to stratify patients, particularly focusing on those with RAS/BRAF wild type mCRC.

pubmed.ncbi.nlm.nih.gov

HIGH Stage IV ctDNA VERIFIED LIQUID BIOPSY PUBLISHED SEP 20, 2026 7

ctDNA in the Management of Resectable & Advanced Colorectal Cancer: Current Status and Future Directions

Circulating tumor DNA (ctDNA) serves as a biomarker for molecular residual disease (MRD) in colorectal cancer management post-curative treatment. This narrative review synthesizes evidence across various stages of CRC, highlighting the need to differentiate ctDNA's prognostic value from its clinical utility. The findings underscore ctDNA's potential role in monitoring and guiding treatment decisions in resectable and advanced CRC.

pubmed.ncbi.nlm.nih.gov

MEDIUM All Stages NGS VERIFIED GENOMIC PROFILING PUBLISHED OCT 01, 2026 5

Distinct NGS mutational landscape and prognostic implications in early-onset colorectal cancer: A dual-cohort analysis of TCGA PanCancer Atlas and MSK-IMPACT 50K

pubmed.ncbi.nlm.nih.gov

MEDIUM Stage III BROAD SCOPE VERIFIED RESEARCH PUBLISHED OCT 08, 2026 4

Epidemiology of resected colorectal liver metastasis and use of perioperative chemotherapy: a prospective cohort study on the impact of next generation sequencing and mutation profiling in a real-world population

pubmed.ncbi.nlm.nih.gov

HIGH Stage III ctDNA VERIFIED LIQUID BIOPSY PUBLISHED OCT 05, 2026 4

Role of circulating tumour DNA in predicting complete clinical response and local regrowth after total neoadjuvant therapy for rectal cancer: meta-analysis

pubmed.ncbi.nlm.nih.gov

HIGH All Stages ctDNA VERIFIED LIQUID BIOPSY PUBLISHED OCT 01, 2026 4

Clinical impact of prospective circulating tumour DNA testing for minimal residual disease in colorectal cancer: the INTERCEPT programme experience

pubmed.ncbi.nlm.nih.gov

MEDIUM All Stages BROAD SCOPE VERIFIED GENOMIC PROFILING PUBLISHED SEP 26, 2026 4

Beyond Simple Ratios: Can the Preoperative HALP Score Provide a More Effective Inflammatory Biomarker for Colon Cancer Survival?

The study investigates the preoperative HALP score as an inflammatory biomarker for predicting overall survival in colon cancer patients undergoing surgery. It compares the HALP score with other inflammatory and nutritional biomarkers to assess their predictive abilities. The findings suggest that the HALP score may offer enhanced prognostic value, although specific outcomes are not detailed in the abstract.

pubmed.ncbi.nlm.nih.gov

MEDIUM All Stages MSI/MMR VERIFIED GENOMIC PROFILING PUBLISHED OCT 01, 2026 3

Age- and sex-specific prevalence of deficient mismatch repair protein expression in gastric, colorectal, and endometrial cancers: a center-based study of older Japanese patients

pubmed.ncbi.nlm.nih.gov

MEDIUM Stage IV MSI/MMR VERIFIED GENOMIC PROFILING PUBLISHED SEP 24, 2026 3

LMNA-NTRK1 fusion-positive MSI-high colorectal cancer responding to FOLFOX plus bevacizumab after immune checkpoint inhibitor failure: a case report

A 63-year-old woman with LMNA-NTRK1 fusion-positive, microsatellite instability-high metastatic colorectal cancer progressed on pembrolizumab. Following this, she received second-line FOLFOX plus bevacizumab, resulting in a complete response maintained for over three years. Comprehensive genomic profiling was conducted during chemotherapy, highlighting the potential for targeted therapy in this context.

pubmed.ncbi.nlm.nih.gov

MEDIUM All Stages MSI/MMR VERIFIED RESEARCH PUBLISHED OCT 07, 2026 2

Pan-cancer prevalence of microsatellite instability and Lynch syndrome in India

pubmed.ncbi.nlm.nih.gov

MEDIUM All Stages BRAF VERIFIED RESEARCH PUBLISHED SEP 29, 2026 2

KRAS and BRAF Hotspot Variants Show Lower Prevalence in Bangladeshi Colorectal Cancer Patients

pubmed.ncbi.nlm.nih.gov

MEDIUM Early Detection NGS VERIFIED GENOMIC PROFILING PUBLISHED SEP 24, 2026 2

A Portable Colorimetric Biosensor Platform for Urinary Colorectal Cancer Biomarker Testing in Low-Resource Settings

A low-cost, semi-automated point-of-care platform for urinary colorectal cancer biomarker testing addresses early detection challenges in low-resource settings. The system utilizes urine-based metabolite biomarkers, providing a non-invasive alternative to traditional colonoscopy and laboratory testing. Key features include standardized fluid handling and quantitative calibration, enhancing accessibility for CRC triage.

pubmed.ncbi.nlm.nih.gov

HIGH All Stages NGS VERIFIED GUIDELINE PUBLISHED SEP 15, 2026 2

Paired Exome-Based Comprehensive Genomic Profiling and Germline Genetic Testing for Unselected Patients With Colorectal Cancer in a Multicenter Prospective Study

Universal combined germline genetic testing (GGT) and comprehensive genomic profiling (CGP) enhances clinical utility for early-onset colorectal cancer (EOC) and elderly-stage colorectal cancer (ESC) patients. This approach accelerates the identification of germline mutations in hereditary cancer genes, improving the timely selection of therapies, clinical trials, and management strategies, particularly for EOC patients who typically have worse outcomes.

pubmed.ncbi.nlm.nih.gov

MEDIUM Early Detection MSI/MMR VERIFIED EARLY DETECTION PUBLISHED SEP 11, 2026 2

A Lynch Syndrome Family With Germline MLH1 c.931A>G Showing Preserved Tumor MMR Immunostaining but MSI-H Status: A Case Report

A family with the germline MLH1 c.931A>G variant exhibited retained mismatch repair protein expression via immunohistochemistry despite showing microsatellite instability-high (MSI-H) status on tumor sequencing. This discordance highlights potential complexities in Lynch syndrome screening and underscores the need for comprehensive testing approaches in colorectal cancer cases.

pubmed.ncbi.nlm.nih.gov

MEDIUM Early Detection Methylation VERIFIED GENOMIC PROFILING PUBLISHED OCT 07, 2026 1

Development of an application for colorectal cancer prediction using SDC2 and TFPI2 methylation: an opportunity for early and noninvasive diagnosis in Latin-America

pubmed.ncbi.nlm.nih.gov

MEDIUM Early Detection BROAD SCOPE VERIFIED GENOMIC PROFILING PUBLISHED OCT 01, 2026 1

Using FIT Thresholds for Priority Triage and Risk Stratification in Resource-Constrained Systems

pubmed.ncbi.nlm.nih.gov

MEDIUM Stage IV BRAF VERIFIED RESEARCH PUBLISHED SEP 26, 2026 1

Systemic inflammation and prognosis in patients with RAS/BRAF wild-type metastatic colorectal cancer receiving anti-EGFR therapy: A real-world cohort study

In patients with RAS/BRAF wild-type metastatic colorectal cancer receiving anti-EGFR therapy, routinely assessed markers of systemic inflammation are independently prognostic. These markers may serve as useful adjuncts for oncologists in discussions regarding patient prognosis.

pubmed.ncbi.nlm.nih.gov

MEDIUM Stage III MSI/MMR VERIFIED RESEARCH PUBLISHED SEP 17, 2026 1

Can Chemoradiotherapy Expand Immune Checkpoint Blockade Responsiveness in Mismatch Repair-Proficient Rectal Cancer?

In mismatch repair-proficient (pMMR) rectal cancer, the addition of immune checkpoint blockade (ICB) to chemoradiotherapy (CRT) has been linked to pathological complete responses in a subset of patients. This suggests that the neoadjuvant context may enhance ICB responsiveness, despite the lack of benefit in advanced pMMR disease. Further investigation into this combination could inform treatment strategies for pMMR rectal cancer.

pubmed.ncbi.nlm.nih.gov

MEDIUM All Stages BROAD SCOPE VERIFIED RESEARCH PUBLISHED SEP 15, 2026 1

Mechanism-Driven Diagnostic Development: A Specimen-Aware Framework Illustrated by Colorectal Cancer and Solid Tumours

A mechanistic, multi-layered, and specimen-aware approach to biomarker testing may enhance cancer detection, classification, prognosis, MRD assessment, and therapeutic selection in colorectal cancer and other solid tumors. The review highlights the limitations of current reductionist biomarker strategies and suggests that integrating complex molecular profiling could better represent cancer biology and improve clinical outcomes.

pubmed.ncbi.nlm.nih.gov

MEDIUM All Stages EGFR VERIFIED RESEARCH PUBLISHED SEP 12, 2026 1

Quantitative dot blot method for measuring EGFR protein levels in patients with colorectal cancer

A quantitative dot blot (QDB) assay was developed to measure EGFR protein levels in colorectal cancer specimens. This method provides an objective assessment of EGFR as a predictive biomarker, facilitating both retrospective and prospective evaluations of anti-EGFR therapies in clinical trials.

pubmed.ncbi.nlm.nih.gov

Appendix A — Patient-Friendly Summary

Plain-language overview for patients, caregivers, and family members

What's happening

Biomarker testing — also called molecular profiling or genomic testing — analyzes your tumor's DNA to identify specific mutations. The results determine which FDA-approved treatments may be available to you. The entries below represent the highest-scoring study per biomarker target drawn from the full research archive — not limited to the current report window. Some entries may predate the timeline above; they are included because they remain the most clinically significant reference available for that biomarker.

ctDNA

Circulating tumor DNA tumor fraction as a biomarker in refractory metastatic colorectal cancer: findings from the CAVE-2 GOIM trial

In the CAVE-2 GOIM trial, circulating tumor DNA (ctDNA) was assessed using the FoundationOne Liquid assay to evaluate its role in guiding anti-EGFR rechallenge therapy with cetuximab ± avelumab in refractory metastatic colorectal cancer patients. The study highlights the need for biomarkers to stratify patients, particularly focusing on those with RAS/BRAF wild type mCRC.

MSI/MMR

EMPIRE (NSABP FC-13): a biomarker-driven phase II platform trial evaluating cemiplimab-based immunotherapy in microsatellite-stable colorectal cancer with ctDNA-defined minimal residual disease

In the EMPIRE (NSABP FC-13) phase II trial, cemiplimab-based immunotherapy was evaluated in microsatellite-stable colorectal cancer patients with ctDNA-defined minimal residual disease. The study aims to identify high-risk patients post-curative therapy who may benefit from additional treatment, addressing the limited efficacy of immune checkpoint inhibitors in this population.

KRAS

Circulating tumor DNA (ctDNA) clearance as an early indicator of sotorasib + panitumumab efficacy and prognosis in KRAS G12C–mutated colorectal cancer (mCRC): Results from phase 3 CodeBreaK 300.

In the phase 3 CodeBreaK 300 trial, ctDNA clearance was assessed as an early indicator of treatment efficacy and prognosis in KRAS G12C–mutated mCRC patients receiving sotorasib and panitumumab. The findings suggest that ctDNA clearance may serve as a valuable biomarker for monitoring therapeutic response and predicting outcomes in this patient population.

TP53

Clinical impact of TP53 mutation status on survival outcomes in metastatic colorectal cancer

TP53 mutations serve as an independent prognostic marker for prolonged progression-free survival in metastatic colorectal cancer. While overall survival differences were not statistically significant, the findings suggest the need for further validation in prospective studies to assess TP53's role in therapeutic stratification.

BRAF

Detection of KRAS, NRAS and BRAF Mutations in Liquid Biopsy from Patients with Colorectal Cancer

ddPCR analysis of circulating tumor DNA (ctDNA) in colorectal cancer patients reveals that it can provide complementary information for molecular diagnosis. The study highlights the utility of liquid biopsy in detecting KRAS, NRAS, and BRAF mutations, enhancing diagnostic accuracy in this patient population.

NGS

Distinct NGS mutational landscape and prognostic implications in early-onset colorectal cancer: A dual-cohort analysis of TCGA PanCancer Atlas and MSK-IMPACT 50K

EGFR

Second-line raltitrexed plus irinotecan (SALIRI) with bevacizumab, cetuximab, or alone for metastatic colorectal cancer: a prospective multicenter cohort-based secondary analysis of the SALLY trial

SALIRI, a combination of raltitrexed and irinotecan, demonstrated feasibility and tolerability as a second-line therapy for metastatic colorectal cancer when combined with bevacizumab or cetuximab in a nonrandomized secondary analysis of the SALLY trial. Favorable outcomes were noted in the SALIRI + cetuximab subgroup, though results should be interpreted with caution due to nonrandom treatment allocation. Further prospective, biomarker-driven randomized studies are needed to clarify the optimal use of SALIRI-based combinations.

RAS

The temporal evaluation of RAS mutation by liquid biopsy at progression after bevacizumab-based treatment in patients with metastatic colorectal cancer

Dynamic changes in plasma RAS mutation status during disease progression in metastatic colorectal cancer (mCRC) were observed, indicating tumor clonal evolution. Notably, RAS mutation clearance correlated with a survival advantage, suggesting prognostic relevance. Reassessing RAS status via liquid biopsy at progression may yield clinically relevant insights, though findings are exploratory and warrant further investigation.

HER2

Comprehensive genomic landscape of ERBB2 in Chinese GI tumors: mutation-centered landscapes and precision treatment opportunities

ERBB2 alterations in colorectal cancer (CRC) and gastric cancer (GC) predominantly occur in kinase domain hotspots, influenced by tumor-specific genomic contexts. Oncogenic ERBB2 mutations correlate with higher tumor mutational burden (TMB) and microsatellite instability-high (MSI-H) status. These findings support the clinical evaluation of mutation-selective HER2 inhibitors and their potential combination with immune checkpoint blockade for targeted therapy in CRC and GC.

Genomic Profiling

Integrated Molecular Profiling of Colorectal Cancer by Tumor Location: Evidence from a Real-World Cohort with Primary and Metastatic Samples

Colorectal cancer (CRC) tumor location correlates with distinct molecular profiles. Right-sided tumors exhibit dMMR, MSI-H, and elevated TMB, while left-sided tumors show increased ERBB2 alterations and class 5 APC mutations. These findings underscore the necessity of incorporating tumor location into personalized molecular diagnostics and treatment strategies for CRC patients.

Methylation

Lactate as a Master Regulator of Immune Suppression: From Metabolic Waste to Epigenetic Checkpoint in Colorectal Cancer

Lactate, a byproduct of aerobic glycolysis driven by KRAS and BRAF mutations in microsatellite-stable colorectal cancer, acts as a central immunosuppressive regulator in the tumor microenvironment. Its accumulation leads to immune suppression, contributing to resistance against immune checkpoint inhibitors. This highlights lactate's role beyond metabolic waste, suggesting potential therapeutic targets to enhance immune response in this cancer subtype.

TMB

Co-Mutation of CREBBP/EP300 and POLE/POLD1 Identifies a TMB-High Subset of MSS CRC

Co-mutation of CREBBP/EP300 and POLE/POLD1 identifies a subset of TMB-high microsatellite stable (MSS) colorectal cancer (CRC). Analysis of multiple public cohorts demonstrated that CREBBP/EP300 mutations are independently associated with TMB-high status ($p < 0.00001$). This finding suggests potential for these mutations as biomarkers for immunotherapy eligibility in MSS CRC.

miRNA

Diagnostic Performance of Circulating miRNA-92a and Related Circulating miRNA Panels for Colorectal Cancer: An Updated Systematic Review and Meta-Analysis

Circulating miRNA-92a shows superior diagnostic performance as a non-invasive biomarker for early colorectal cancer detection. The systematic review and meta-analysis suggest its potential utility, though further large-scale prospective studies are needed to standardize testing protocols and confirm clinical applicability.

Stool DNA

Enhancing non-invasive colorectal cancer screening with stool DNA methylation markers and light gradient-boosting machine learning

Stool DNA methylation markers CNRIP1, SFRP2, and VIM demonstrated high sensitivity for non-invasive colorectal cancer screening. The application of the LightGBM machine learning algorithm improved diagnostic accuracy, suggesting a viable method for early CRC detection.

IHC

Integration of deep learning and radiomic features from multiplex immunohistochemistry images for reproducible Multi-Outcome prediction in a Multi-Center study of colorectal cancer

Fused multiplex immunohistochemistry (mIHC)-derived radiomic and deep learning features provide accurate and interpretable predictions for multiple colorectal cancer outcomes. The study supports the integration of these features into precision oncology workflows, enhancing the potential for reproducible multi-outcome predictions across different centers.

What this means for you

If you have been diagnosed with colorectal cancer, ask your oncologist: *"Has my tumor been tested for all the biomarkers that have an FDA-approved treatment?"* This is especially important if you have been diagnosed with metastatic (Stage IV) colorectal cancer, where biomarker results directly determine which targeted therapies and immunotherapies are available to you. Based on the research in this report, the most actively studied biomarkers right now are **ctDNA, MSI/MMR, KRAS, TP53, and BRAF**. Testing should happen at diagnosis — before treatment begins — because results take 1–2 weeks and can shape your first treatment decision.

Generated from structured biomarker research data · 2026-10-11 07:30:01
This is not medical advice. Always speak with your oncologist before making any treatment decisions.

Appendix B — Colon & Colorectal Cancer: Biomarker Testing Reference

What Is Biomarker Testing?

Biomarker testing — also called molecular profiling or genomic testing — analyzes your tumor's DNA to identify specific mutations. A single next-generation sequencing (NGS) panel can check many gene mutations at once. Results typically take 1–2 weeks and should be obtained at diagnosis, before treatment begins.

Key Biomarkers — Test All mCRC at Diagnosis

RAS (KRAS/NRAS) · BRAF V600E · MSI/MMR status · HER2 · NTRK fusion · TMB

Testing Methods

NGS (next-generation sequencing) — checks many mutations at once from tumor tissue

Liquid biopsy / ctDNA — blood test that detects tumor DNA; useful when tissue is unavailable

IHC (immunohistochemistry) — used to test MMR status and HER2

PCR — used for specific mutation detection (e.g. MSI, KRAS)

Questions to Ask Your Oncologist

"Has my tumor been tested for all the biomarkers that have an FDA-approved treatment?"

"Was my sample taken from the primary tumor or a metastatic site — and does it need to be retested?"

"How long will results take, and will they affect my first treatment decision?"

"Is a liquid biopsy appropriate for my situation?"

Abbreviations

BRAF V600E — a specific gene mutation found in ~8–10% of colorectal cancers

ctDNA — circulating tumor DNA; tumor DNA fragments found in the bloodstream

dMMR — mismatch repair deficient; indicates the tumor may respond to immunotherapy

HER2 — a protein that drives growth in ~2–3% of colorectal cancers

IHC — immunohistochemistry; a lab staining method used to detect proteins in tissue

KRAS G12C — a specific gene mutation found in ~3–4% of colorectal cancers

mCRC — metastatic colorectal cancer (Stage IV)

MSI-H — microsatellite instability-high; indicates the tumor may respond to immunotherapy

NGS — next-generation sequencing; a method that checks many gene mutations at once

NTRK — a rare gene fusion found in <1% of colorectal cancers

RAS — a gene family (KRAS/NRAS) whose mutation status affects treatment selection

TMB — tumor mutational burden; a measure of how many mutations are in a tumor

Key Resources

ClinicalTrials.gov · NCCN Guidelines (nccn.org) · Colorectal Cancer Alliance (ccalliance.org) · Fight Colorectal Cancer (fightcrc.org) · ACS (cancer.org)

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About This Report

This report is generated automatically from publicly available medical databases including PubMed, ASCO Journal of Clinical Oncology, Annals of Oncology, and NCI Cancer.gov. Entries are classified using deterministic rule-based logic and scored by clinical significance. AI-generated summaries are grounded exclusively in the source data provided.

Limitations

This report does not represent a complete picture of all published research. Database coverage, feed availability, and relevance filtering affect what appears. Research findings may not yet reflect current clinical guidelines. Always verify information at the original source before making clinical decisions.

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